

PREPARED FOR

T6N, R4W, Mount Diablo Meridian

COORDINATES 38.3459°N 122.3047°W WGS84	ELEVATION 66 ft NAVD88 · MSL	STATE PLANE CA 2 (0402) · N 1,887,922.81 E 6,474,289.05 (US ft, NAD83) -	FEMA ZONE Zone X 06055C0510FF	DECLINATION 13.0° E WMM 2026
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APN **038050009000** | ZONING **AP** | ACRES **0.481 ac**

BEST GNSS WINDOW **Mon, Jul 13 · 6:00 AM – 10:00 AM** | PDOP **0.99 · 34 sv** | NEAREST CORS
P261 · 14.1 mi ✓

REPORT CONTENTS

01	7-DAY GNSS FIELD RANKING	202	SITE & SIGNAL REPORT	3
03	BEST STATIC OBS WINDOW	404	EMERGENCY HOSPITALS	5
05	DAY-BY-DAY WEATHER + DRONE	606	AI FIELD ANALYSIS	13
07	WATER LEVELS & GAUGES	1408	FAULTS & SUBSIDENCE	15
09	STRUCTURAL PARAMETERS	1610	SOILS & ENGINEERING	16
11	MINES & MINERALS	1712	NATIVE PLANTS & WILDLIFE	18
13	FCC MICROWAVE PATHS	1914	SOLAR & PASSIVE DESIGN	21
15	AI SITE INTELLIGENCE BRIEF	2316	FIELD MAP · TOPO · DXF SITE PLAN	24



Site file · Google Earth
Download KMZ

RECORDER
Napa County Recorder-Clerk · 1127 First
St, Napa, CA 94559

NEAREST CORS
P261 · 14.1 mi

GENERATED
July 13, 2026

ASSESSOR
Napa County Assessor · 1127 First St,
Napa, CA 94559

NEAREST BM
JT9010 · 0.2 mi

SYSTEM
TopoScout · info@toposcout.com

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

SITE & SIGNAL REPORT

COORDINATES

38.3459, -122.3047

WGS84 / NAD83

ELEVATION

66 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.0° E

Compass reads 13.0° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06055C0510FF

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.4

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

23.9 TECU

+0.3 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.4). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Mon, Jul 13	0.99	6:00 AM – 10:00 AM	34	Ideal

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates. For site-specific weather, terrain mask angle, and nearest CORS station, order an On-Demand Site Report.



WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 13, 2026) · Conditions may change – verify before fieldwork

✖ GPS CONSTELLATION HEALTH

▲ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–0.8 across survey window
PDOP remains ≤ 0.8 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 92°F | Low 60°F | Wind max 12 mph | Gusts 14 mph | Precip 0.00" | Sunrise 5:56 AM | Sunset 8:33 PM

Waning Crescent 2%

GNSS: Avg PDOP 0.99 · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.84	36 sv	IDEAL
7:00 AM	0.95	34 sv	IDEAL
8:00 AM	0.95	33 sv	IDEAL
9:00 AM	1.00	33 sv	GOOD
10:00 AM	1.01	29 sv	GOOD
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.03	26 sv	GOOD
1:00 PM	1.08	26 sv	GOOD
2:00 PM	0.93	32 sv	IDEAL
3:00 PM	0.97	33 sv	IDEAL
4:00 PM	0.98	33 sv	IDEAL
5:00 PM	0.99	33 sv	IDEAL
6:00 PM	1.02	30 sv	GOOD
7:00 PM	0.99	32 sv	IDEAL



GO - GOOD CONDITIONS

Partly cloudy | Wind: 11.6 mph (gusts 14.1 mph) | Precip: 0" | Temp: 59.6°–92°F



DRONE WIND ANALYSIS



GO Peak: 11.6/14.1 mph S

* Sunrise: 5:56 AM | Sunset: 8:33 PM | fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	1/3 mph		ESE
6 AM	1/2 mph		NW
7 AM	1/1 mph		WNW
8 AM	2/3 mph		SSE
9 AM	3/4 mph		S
10 AM	3/4 mph		SSE
11 AM	6/7 mph		S
12 PM	8/10 mph		S
1 PM	8/9 mph		S
2 PM	9/11 mph		S
3 PM	9/11 mph		S
4 PM	12/14 mph		S
5 PM	8/13 mph		SSW
6 PM	10/12 mph		S
7 PM	9/11 mph		SSE
8 PM	5/8 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

GNSS SURVEY FIELD PLANNING REPORT

Site: 38.3459, -122.3047 (Napa County, CA)

Survey Date: Monday, July 13, 2026

Report Generated: Monday, July 13, 2026

VERDICT

GO — Favorable PDOP geometry (0.94–0.99) and acceptable weather support survey execution on the scheduled date.

BEST SURVEY WINDOW

Monday, July 13, 2026 | 06:00 AM – 10:00 AM (Pacific)

RATIONALE: PDOP 0.94 at peak (6:00–10:00 AM window) with 34 visible satellites (GPS+GLONASS+Galileo+BeiDou, 15° mask). Multi-constellation availability ensures robust geometry despite GPS PRN 11 unhealthy status. Receivers will auto-exclude PRN 11; constellation degradation is negligible in this window.

BEST 4-HOUR STATIC GNSS SESSION

Conduct primary static occupation: 06:30 AM – 10:30 AM

Rationale: Captures peak PDOP improvement (0.94) for minimum 4 hours during high-satellite-count period. Moderate ionospheric activity (Kp 4.7, TEC +0.3 TECU) is manageable; longer occupation times (4+ hours) mitigate residual ionospheric delay. Post-process with multi-constellation baseline solutions.

DRONE OPERATIONS ASSESSMENT

Weather: ACCEPTABLE (overcast, 12 mph max wind, zero precipitation). Ceiling unspecified but overcast condition suggests safe visual line-of-sight operations within site parameters. Wind within typical operational limits.

Air Quality: GOOD (AQI 45). No respiratory/visibility constraints.

Land Cover: Dwarf scrub with low sky obstruction. Drone takeoff and RTK base siting feasible. Confirm local airspace restrictions (Napa County proximity to wine-country airspace); no wildfire TFR active.

Recommendation: Proceed with planned drone surveys during 06:00–10:00 AM window to coincide with optimal GNSS geometry.

FIELD SAFETY WARNINGS

- GPS PRN 11 (UNUSUFN) is unhealthy with no return-to-service estimate. No action required; receivers auto-exclude. Document in survey notes for QA.
- Moderate geomagnetic activity (Kp 4.7). Ionospheric delay may introduce 5–15 cm range errors in standalone positioning; acceptable for static post-processing with multi-constellation fixes. Recommend VRS or RTK baselines if real-time cm-level accuracy required.
- TEC data flagged quality_flag non-zero (estimate only); use for planning context only. Do not rely for differential correction thresholds.
- Overcast conditions reduce direct solar exposure; crew heat stress risk is low. Standard hydration protocol sufficient.

SUMMARY

Monday July 13, 2026 is CLEAR FOR OPERATIONS. PDOP geometry is excellent (0.94 best, 0.99 average) with 34-satellite multi-constellation constellation; moderate ionospheric activity (Kp 4.7, +0.3 TECU) is routine and managed by 4+ hour static sessions and post-processing. Weather is favorable (overcast, 12 mph winds, no precipitation). GPS PRN 11 unhealthy status has negligible impact on constellation geometry. Conduct primary static GNSS session 06:30–10:30 AM to maximize PDOP and satellite count; synchronize drone RTK/PPK operations to this window. No safety constraints identified. Proceed with full survey plan as scheduled.

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Napa County Recorder-Clerk

1127 First St, Napa, CA 94559

ASSESSOR / TAX OFFICE

Napa County Assessor

1127 First St, Napa, CA 94559

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P261	HUNTERHILLCN2004	14.1 mi
 NGS ↗	P198	PETALUMAIRCN2004	17.5 mi
 NGS ↗	OHLN	OHLONE PARK	23.5 mi
 NGS ↗	P196	MEACHUMLFLCN2006	24.0 mi
 NGS ↗	CASR	SANTA ROSA CA	24.9 mi
 NGS ↗	P267	DIXONAVIATCN2005	26.3 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	JT9010	"13"	19.100m / 62.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9018	"13 E"	17.970m / 59.0ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9009	"13"	17.900m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9017	"13 D"	17.890m / 58.7ft NAVD88	Order	0.2 mi
 Datasheet ↗	JT9034	"20 A"	20.730m / 68.0ft NAVD88	Order	0.3 mi
 Datasheet ↗	JT9012	"13"	19.060m / 62.5ft NAVD88	Order	0.4 mi
 Datasheet ↗	JT9043	"25 A"	18.710m / 61.4ft NAVD88	Order	0.4 mi
	JT9016	"13 D"	18.200m / 59.7ft NAVD88	Order	0.5 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet ↗					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06055C0510FF

Effective: Sep 29, 2010

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
JT9010	13	38.348333	-122.302778	62.7	
JT9018	13 E	38.348333	-122.302778	59.0	
JT9009	13	38.346667	-122.301111	58.7	
JT9017	13 D	38.345000	-122.301111	58.7	
JT9034	20 A	38.345000	-122.309444	68.0	
JT9012	13	38.351667	-122.304444	62.5	
JT9043	25 A	38.340000	-122.302778	61.4	
JT9016	13 D	38.341667	-122.297778	59.7	

Download .txt File

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	60.6 mi	696.94 ft (-9 ft)	706 ft	36K af	67%
Folsom Lake	66.5 mi	448.22 ft (-18 ft)	466 ft	788K af	81%
Lake Oroville	93.4 mi	861.25 ft (-39 ft)	900 ft	2,859K af	81%
New Melones Lake	100.1 mi	1,020.22 ft (-68 ft)	1,088 ft	1,664K af	69%
Black Butte Lake	101.2 mi	—	228 ft	74K af	54%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
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ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	64 mi	696.9 ft	695.01 ft	—	100%
Camanche	89 mi	230.3 ft	221.86 ft	—	104%
Englebright Lake-Pool	95 mi	525.1 ft	522.35 ft	—	101%
Farmington Dam-Pool	99 mi	124.9 ft	117.09 ft	—	107%
Black Butte-Pool	101 mi	457 ft	458.36 ft	—	100%
Don Pedro	137 mi	—	N/R	—	—

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
005N004W32A002M	7 mi	200 ft	87.71 ft BGS	2014-11-05	N100CACSTL
005N004W28R001M	7 mi	51 ft	41.90 ft BGS	1977-10-03	N100CACSTL
004N004W04C001M	8 mi	80 ft	11.20 ft BGS	1977-10-03	N100CACSTL
004N004W05B001M	8 mi	—	54.10 ft BGS	1977-10-03	N100CACSTL
005N004W31R001M	8.1 mi	295 ft	—	—	—
004N004W05D002M	8.3 mi	60 ft	14.60 ft BGS	1977-10-03	N100CACSTL
004N004W06B001M	8.3 mi	201 ft	21.86 ft BGS	2022-06-07	—
004N004W02L001M	8.8 mi	—	9.90 ft BGS	1977-10-02	N100CACSTL

🕒 AI AQUIFER INTELLIGENCE — CLAUDE HAIKU

The Napa Sonoma Coast Ranges aquifer system (code N100CACSTL) consists of fractured bedrock and alluvial deposits that formed during the Tertiary and Quaternary periods through tectonic uplift and erosion of coastal mountain terrain. Water in this system is stored in fractures and limited intergranular spaces within consolidated rock and unconsolidated deposits, making yields highly variable and dependent on local fracture connectivity. At this Napa County site, water availability is moderate but sensitive to both seasonal recharge and longer-term basin stress.

FORMATION

Name	Napa Sonoma Coast Ranges (Cascadia/Sierran Tertiary, local Quaternary alluvium)
Age / Rock	Tertiary to Quaternary fractured metamorphic and sedimentary bedrock with Quaternary alluvial cover.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	51–295 ft (5–30 floors)
Typical Depth	80–120 ft (8–12 floors)
Recharge Source	Precipitation on ridge and hillslope outcrops plus lateral flow through fractured bedrock and alluvial aquifer margins
Transit Time	20–50 years to 100 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.8–7.4 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)

Notes	No sample data available for this site. Regional water is typically soft to moderately hard with low salinity; local iron and manganese may occur in reducing zones.
TREND	
Trend	Significant long-term decline: water levels dropped from 9.90–54.10 ft BGS (1977) to 21.86–87.71 ft BGS (2014–2022), reflecting cumulative groundwater abstraction and sustained drought.
Source	Site well records (1977, 2014, 2022) and regional USGS/DWR trends
Seasonal Swing	est. 8–18 ft (est. USGS/DWR regional data)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**
Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU	
The study area near Sonoma County (38.3459, -122.3047) overlies the N100CACSTL aquifer system in a region with well-drained soils and is NOT within a critically overdrafted groundwater basin as defined by SGMA. The geological setting features consolidated Tertiary and older bedrock with limited clay-rich compressible sequences typical of subsidence-prone areas, and groundwater extraction has not induced significant aquifer system compaction. Surveyors working in this area can proceed with standard elevation verification protocols without subsidence-specific adjustments to benchmark reliability.	
SUBSIDENCE ASSESSMENT	
Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin
BENCHMARK IMPLICATIONS	
Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate
SURVEY ACTION	
Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)		WIND / SNOW / ICE / FROST (ASCE 7-22)	
Site Class	D (default)	Basic Wind Speed	92 mph
Risk Category	II	Ground Snow Load	5.8 psf
		Winter Wind (W2)	0.35
		Ice Thickness	None
		Frost Depth	12 in minimum
		Est. from IECC Zone 3C — AFI data unavailable Verify with local building official before construction	
		Source: ASCE 7-22 Hazard Tool · gis.asce.org	

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Napa

🕒 **AI CLIMATE SUMMARY — RESIDENTIAL DESIGN**
This marine climate demands a well-sealed, moderately insulated envelope that can handle occasional high winds—focus on continuous air barriers and impact-resistant glazing, particularly on the ocean-facing or windward side. Concentrate glazing on the south and east exposures to capture winter sun while minimizing west-facing glass, which will overheat during mild afternoons and increase cooling loads. The key passive strategy here is cross-ventilation: design operable windows on opposite sides of living spaces to let cool coastal breezes naturally condition the home during the frequent mild days, reducing mechanical cooling needs year-round.

🕒 AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU	
This coastal marine site in IECC Zone 3C presents moderate wind exposure and light snow loads with well-drained gravelly loam soils. Seismic parameters are unavailable (null values) and may warrant immediate clarification with the local building official; assuming Site Class D for planning purposes suggests moderate foundation design considerations pending confirmation. Conventional spread footings may be anticipated, though a geotechnical investigation is suggested to verify soil bearing capacity and confirm seismic classification.	
FOUNDATION	
Type	Conventional spread footing anticipated pending geotechnical investigation
Basis	Well-drained gravelly loam (Coombs series) with 2–5% slopes suggests adequate bearing capacity for standard foundations; however, seismic parameters are null and Site Class D is assumed for planning—consult a licensed engineer to confirm SDC and adjust detailing accordingly
STRUCTURAL IMPLICATIONS	
Seismic	Seismic Design Category cannot be determined from available data (Ss, S1, SDS, SD1 all null). SDC A–B ordinary detailing may be anticipated for this coastal location, but confirmation with local building official is essential before final design

Wind Exposure	Exposure C (open terrain) suggested per ASCE 7-22 Risk Category II with 92 mph 3-second gust anticipated
Wind	Standard wind load design anticipated—coastal marine climate may warrant verification of exposure category and consideration of wind-borne debris protection depending on proximity to ocean
Snow	Light snow load of 5.8 psf anticipated with winter wind factor W2=0.35—standard roof design likely adequate; consult engineer for any drift or exposure concerns specific to final building geometry
INVESTIGATIONS & INSPECTIONS	
Geotech Required	Yes—geotechnical investigation recommended to confirm Site Class D, bearing capacity, and any seismic implications; frost depth estimate of 12 in should be verified with local building official
Liquefaction Screen	Screening recommended if seismic classification is elevated—groundwater depth and soil saturation conditions should be evaluated; low risk may be anticipated given well-drained soil classification and non-subsidence status
Special Inspections	May be required if SDC is determined to be C or higher—consult engineer once seismic parameters are clarified
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning purposes. Geotechnical investigation strongly recommended to confirm soil profile, bearing capacity, and seismic site class; clarify null seismic parameters with local building official immediately.

SOILS — USDA SOIL SURVEY

MAP UNIT: Coombs gravelly loam, 2 to 5 percent slopes	SERIES: Coombs
DRAINAGE CLASS: Well drained	HYDRO GROUP: C
SOIL ORDER: Alfisols	SUBORDER: Xeralfs

○ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Coombs gravelly loam is a well-drained Alfisol (Xeralf) formed in mixed alluvial and residual materials on gently rolling terrain in the Coast Ranges. The soil profile features a gravelly loam surface layer overlying a clay loam argillic horizon, typical of Mediterranean climates with seasonal water movement. Surveyors and developers should expect moderate gravel content affecting excavation efficiency, good bearing capacity for shallow foundations, and minimal drainage concerns on these 2–5% slopes.

Mixed alluvial and residual parent material; foothill and terrace landscape positions; coastal scrub and oak woodland vegetation.

DRAINAGE	
Class	Well drained
Seasonal Saturation	None within 60 inches
Flooding Risk	Minimal on 2–5% slopes
Ponding Risk	None
ENGINEERING	
Bearing Capacity	3,500–4,500 psf (shallow foundations)
Rating	Good
Shrink-Swell	Moderate — clay loam substratum with seasonal wetting and drying cycles
Frost Action	Low
Cut Suitability	Good (gravel-rich material, stable slopes)
Fill Suitability	Fair to Good (clay loam substratum — moisture control required during compaction)
SEPTIC	
Suitability	Conditional
Notes	Perc test required — clay loam argillic horizon (15–24 inches) may restrict percolation; drainfield design should account for seasonal saturation patterns.
HAZARDS	
Excavation	Moderate — 15–30% gravel content; hand excavation may be necessary in localized patches
Expansive Clay	Low to Moderate — clay content in substratum may cause minor movement under extreme moisture conditions
Erosion	Low on 2–5% slopes; gully risk in disturbed areas with concentrated flow
Other	None

Data sources: California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdnldataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 13, 2026, 1:06 AM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

OIL & GAS WELLS California — within 10mi radius

8 total
0 active
7 plugged
 By status:
 7 Plugged
 1 Idle

By type: 8 DH

Lease / Well	Status	Type	Operator	Depth	Dist
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	7.2 mi
McKenzie-Hazard Syndicate	Plugged	DH	McKenzie-Hazard Syndicate	—	7.4 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.6 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	8.8 mi
Mobil Oil Corporation	Plugged	DH	Mobil Oil Corporation	—	9.5 mi
Capell Oil Corp.	Idle	DH	Capell Oil Corp.	—	9.7 mi
Cordelia Oil & Gas Co.	Plugged	DH	Cordelia Oil & Gas Co.	—	12 mi

AI Assessment — Well Proximity

Well proximity is not a planning concern for this site. The nearest well is a plugged Mobil Oil Corporation dry hole located 7.2 miles away, well beyond any threshold requiring foundation review, environmental assessment, or surface rights investigation. No active producers, injection wells, or other well types exist within actionable distance.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200 **10** **108**
total in search area within 5 mi within 25 mi

Site Name	Status	Commodities	Distance
Roderick Sand and Gravel Pit	Occurrence	Sand/Gravel	1.6 mi
Basalt Rock Co	Producer	Clay	2 mi
Cicero Pumice Quarry	Occurrence	PUM	2.9 mi
Mount George King Pumice Deposit.	Occurrence	PUM	2.9 mi
Basalt Rock Co.	Occurrence	DIT	3.1 mi
Mt George Pumice Deposit	Producer	PUM	3.1 mi
Wilson Pumice Quarry	Occurrence	PUM	3.7 mi
Unknown	Occurrence	Mercury	4.4 mi
Silva Pumice Quarry	Occurrence	PUM	4.7 mi
Silva Pumice Quarry	Producer	PUM	5 mi
Walker Pumice Quarry	Occurrence	PUM	5.1 mi
Walker Pumice Quarry	Producer	PUM	5.1 mi
Mountain	Occurrence	Mercury	5.4 mi
Unknown	Occurrence	Mercury	7.6 mi
Unknown	Occurrence	Stone/Crushed	7.8 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Location	Past Producer	Sand/Gravel	8.5 mi
Unnamed Prospect	Occurrence	Chromium	8.8 mi
Unnamed Location	Occurrence	Chromium	9 mi
Unknown	Occurrence	Chromium	9.1 mi
Bella Oak	Past Producer	Mercury	9.4 mi
Unknown	Occurrence	MG	9.7 mi
Serres Pit	Past Producer	Sand/Gravel	10.4 mi
Unnamed Location	Unknown	Sand/Gravel	10.5 mi
Conn Valley	Occurrence	Manganese	10.6 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

Three active mineral producers operate within 5 miles of the site. Basalt Rock Co (2 miles, clay) and Mt George Pumice Deposit (3.1 miles, pumice) and Silva Pumice Quarry (5 miles, pumice) should be evaluated for potential impacts including heavy truck traffic on access roads during operating hours, dust generation, and noise from crushing or blasting operations. No historical deposits warrant Phase 1 ESA consideration based on proximity criteria.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6 **6** **12**
Endangered Threatened Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
California tiger Salamander	<i>Ambystoma californiense</i>	Endangered
Conservancy fairy shrimp	<i>Branchinecta conservatio</i>	Endangered
Contra Costa goldfields	<i>Lasthenia conjugens</i>	Endangered
Soft bird's-beak	<i>Cordylanthus mollis</i> ssp. <i>mollis</i>	Endangered
Suisun thistle	<i>Cirsium hydrophilum</i> var. <i>hydrophilum</i>	Endangered
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
Alameda whipsnake (=striped racer)	<i>Masticophis lateralis euryxanthus</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
Delta smelt	<i>Hypomesus transpacificus</i>	Threatened
Northern spotted owl	<i>Strix occidentalis caurina</i>	Threatened
Vernal pool fairy shrimp	<i>Branchinecta lynchi</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Assessment – 38.3459, -122.3047 **ESA Section 7 Triggers & Planning Relevance:** This site's proximity to critical habitat for 12 species triggers mandatory ESA Section 7 consultation if federal permits, funding, or agency approvals are required. The most planning-relevant species are California tiger salamander and vernal pool species (tadpole shrimp, fairy shrimp), which typically indicate seasonal wetland/grassland habitats that may constrain development footprints, and California red-legged frog, which often requires habitat connectivity and hydrological protections. The site's location suggests exposure to Suisun Marsh or Central Valley vernal pool complexes—ecosystem types with stringent Section 7 requirements and limited mitigation flexibility. **Recommended Next Step:** Conduct Phase 1 biological reconnaissance to map on-site and adjacent wetland/pool features, grassland conditions, and species-specific habitat elements before design finalization. Simultaneously, confirm federal nexus requirements and initiate pre-consultation coordination with USFWS to clarify critical habitat overlap and potential Incidental Take Permit pathways, which will directly inform site layout, grading limits, and phasing strategy. --- *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

57	5051	2026-07-04
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American black nightshade	<i>Solanum americanum</i>	2026-05-29
Bearded Clover	<i>Trifolium barbigerum</i>	2026-05-15
bird's foot cliffbrake	<i>Pellaea mucronata</i>	2026-04-28
Blow Wives	<i>Achyrachaena mollis</i>	2026-05-20
blue oak	<i>Quercus douglasii</i>	2026-05-09
Bolander's Phacelia	<i>Phacelia bolanderi</i>	2026-04-27
box elder	<i>Acer negundo</i>	2026-05-18
Braun's Giant Horsetail	<i>Equisetum telmateia braunii</i>	2026-06-22
California bay	<i>Umbellularia californica</i>	2026-06-24
California beeplant	<i>Scrophularia californica</i>	2026-05-20
California black oak	<i>Quercus kelloggii</i>	2026-05-15
California buckeye	<i>Aesculus californica</i>	2026-06-02
California Dutchman's Pipe	<i>Aristolochia californica</i>	2026-05-26
California milkwort	<i>Rhinotropis californica</i>	2026-05-15
California mugwort	<i>Artemisia douglasiana</i>	2026-06-02
California poppy	<i>Eschscholzia californica</i>	2026-06-24
California Wild Rose	<i>Rosa californica</i>	2026-05-09
Canyon liveforever	<i>Dudleya cymosa</i>	2026-05-30
-	<i>Castilleja densiflora densiflora</i>	2026-05-09
clay mariposa lily	<i>Calochortus argillosus</i>	2026-05-15
coast live oak	<i>Quercus agrifolia</i>	2026-05-13
coastal woodfern	<i>Dryopteris arguta</i>	2026-05-20
common selfheal	<i>Prunella vulgaris</i>	2026-06-22
Diogenes' lantern	<i>Calochortus amabilis</i>	2026-05-15
distant phacelia	<i>Phacelia distans</i>	2026-05-15
-	<i>Downingia concolor concolor</i>	2026-05-15
Dwarf Brodiaea	<i>Brodiaea terrestris</i>	2026-05-15
Elegant Clarkia	<i>Clarkia unguiculata</i>	2026-05-16

COMMON NAME	SCIENTIFIC NAME	LAST OBS
goldback fern	<i>Pentagramma triangularis</i>	2026-05-20
Ithuriel's Spear	<i>Triteleia laxa</i>	2026-06-15
maroonspot calicoflower	<i>Downingia concolor</i>	2026-05-15
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2026-06-25
narrowleaf mule-ears	<i>Wyethia angustifolia</i>	2026-05-15
Narrowleaf Willow	<i>Salix exigua</i>	2026-06-24
needleleaf navarretia	<i>Navarretia intertexta</i>	2026-05-15
northern California black walnut	<i>Juglans hindsii</i>	2026-06-24
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-05-29
Oregon Ash	<i>Fraxinus latifolia</i>	2026-06-02
Pacific Bleeding Heart	<i>Dicentra formosa</i>	2026-05-04
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-04
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-06-02
pineapple-weed	<i>Matricaria discoidea</i>	2026-05-20
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-05-28
small-headed clover	<i>Trifolium microcephalum</i>	2026-05-15
Superb Mariposa Lily	<i>Calochortus superbus</i>	2026-06-11
thimble clover	<i>Trifolium microdon</i>	2026-05-15
tomcat clover	<i>Trifolium willdenovii</i>	2026-05-15
Toyon	<i>Heteromeles arbutifolia</i>	2026-06-19
trailing blackberry	<i>Rubus ursinus</i>	2026-06-24
Tree Clover	<i>Trifolium ciliolatum</i>	2026-05-15
valley oak	<i>Quercus lobata</i>	2026-06-24
wavy-leafed soap plant	<i>Chlorogalum pomeridianum</i>	2026-05-20
western sycamore	<i>Platanus racemosa</i>	2026-05-17
white brodiaea	<i>Triteleia hyacinthina</i>	2026-05-15
white-tipped clover	<i>Trifolium variegatum</i>	2026-05-15
Winecup Clarkia	<i>Clarkia purpurea</i>	2026-05-15
Yellow Mariposa Lily	<i>Calochortus luteus</i>	2026-06-11

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 38.3459, -122.3047 This site supports a **mixed oak woodland and grassland community**, dominated by blue oak and California black oak with understory species like California buckeye, California bay, and diverse native forbs and clover. The presence of clay mariposa lily, Canyon liveforever, and other drought-tolerant species indicates **well-drained upland terrain with seasonal moisture availability**, typical of interior California foothills with Mediterranean climate patterns. The clay soil indicators (clay mariposa lily preference) combined with spring-blooming ephemerals suggest **clay-rich soils with winter/spring saturation that dry significantly by summer**—grading activities should account for poor drainage during wet season and potential slope stability issues in clay layers. Drainage plans should preserve existing oak canopy and seasonal seep areas where species like Braun's Giant Horsetail establish. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

635	154	91	1439
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		635 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		154 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		91 obs	
Dasymutilla californica		91 obs	
Dasymutilla sackenii		91 obs	
TOXIC PLANTS			

SPECIES	OBSERVATIONS
<i>Toxicodendron diversilobum</i>	1100 obs
<i>Datura wrightii</i>	57 obs
<i>Datura stramonium</i>	57 obs
<i>Datura ferox</i>	57 obs
<i>Conium maculatum</i>	199 obs
<i>Urtica gracilis</i>	83 obs
<i>Urtica gracilis holosericea</i>	83 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area **Highest-Priority Hazards:** Toxic plants dominate this area with nearly 1,500 observations, particularly poison oak (extremely common) and datura species (highly toxic if ingested). Northern Pacific rattlesnakes are the primary animal hazard, with 635 documented sightings—they're present year-round but most active spring through fall. Black widow spiders and velvet ants are secondary concerns but cause significant pain. **Practical Avoidance:** Wear long pants, closed-toe boots, and long sleeves to protect against plant contact and snake bite. Learn to identify poison oak by its three-leaflet clusters and avoid touching it; wash exposed skin immediately after fieldwork. Stay alert for rattlesnakes in tall grass and rocky areas—watch where you step and place your hands, and never approach or handle snakes. Use a headlamp at dawn/dusk to improve visibility. **Seasonal Considerations:** Rattlesnakes peak in activity April through October; exercise heightened vigilance during these months. Poison oak is year-round but causes worse reactions during active growth (spring/summer). Winter work carries lower envenomation risk but poison oak remains a concern. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FCC MICROWAVE PATHS — Licensed beam paths crossing within 1320ft of site

3

545ft

3

Beam Paths

Closest Beam

Fixed P2P

TX CALL	RX CALL	TYPE	PATH MI	CLOSEST
WBO61	WBO59	Fixed Point-to-Point	—	545ft
KYH37	WQAT627	Fixed Point-to-Point	—	927ft
WRCR329	WQAT627	Fixed Point-to-Point	—	1289ft

Source: FCC ULS microwave license database — PostGIS spatial query — Paths within 1320ft (0.25mi) of site — Towers may be 50+ miles away — FCC Part 101 licensed paths

FAA AIRWAYS — ATS Routes crossing site area

26

13

2

11

Total Airways

Jet Routes

Victor Airways

Other Routes

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	317.1°	135.4°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.0°	127.3°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	161.4°	341.7°
T257	T-Route	Low	—	—	321.1°	140.5°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°

AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

8
Rail Lines

OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION
CFNR	Other	1	No	No	—
CFNR	Yard	1	No	No	—
CFNR	Mainline	1	No	No	VALLEJO
SMRT	Mainline	1	No	No	BRAZOS JCT.
NVRR	Mainline	1	No	No	MARTINEZ
USG	Other	1	No	No	—
CFNR	S	1	No	No	—
NVRR	Yard	1	No	No	—

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

2009655146
Total ObservationsTotal SpeciesBirdsMammalsReptilesAmphibians

BIRDS (65 species)

COMMON NAME	Scientific Name	OBS
Wild Turkey	<i>Meleagris gallopavo</i>	10
Snowy Egret	<i>Egretta thula</i>	7
California Quail	<i>Callipepla californica</i>	5
Western Bluebird	<i>Sialia mexicana</i>	5
California Towhee	<i>Melozone crissalis</i>	5
Great Egret	<i>Ardea alba</i>	4
Red-tailed Hawk	<i>Buteo jamaicensis</i>	4
Black Phoebe	<i>Sayornis nigricans</i>	4
Cooper's Hawk	<i>Astur cooperii</i>	3
Ash-throated Flycatcher	<i>Myiarchus cinerascens</i>	3
Western Screech-Owl	<i>Megascops kennicottii</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Mallard	<i>Anas platyrhynchos</i>	3
Bushtit	<i>Psaltriparus minimus</i>	3
Canada Goose	<i>Branta canadensis</i>	3
+ 50 additional species		

MAMMALS (5 species)

COMMON NAME	Scientific Name	OBS
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
Coyote	<i>Canis latrans</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Eastern Fox Squirrel	<i>Sciurus niger</i>	1
Striped Skunk	<i>Mephitis mephitis</i>	1

REPTILES (14 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	10
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	7
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	6
California King Snake	<i>Lampropeltis californiae</i>	3
Pond Slider	<i>Trachemys scripta</i>	2

COMMON NAME	Scientific Name	OBS
Western Skink	<i>Plestiodon skiltonianus</i>	2
Ring-necked Snake	<i>Diadophis punctatus</i>	1
Common Garter Snake	<i>Thamnophis sirtalis</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Pacific Ringneck Snake	<i>Diadophis punctatus amabilis</i>	1
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	1
Common Sharp-tailed Snake	<i>Contia tenuis</i>	1
North American Racer	<i>Coluber constrictor</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1

AMPHIBIANS (6 species)

COMMON NAME	Scientific Name	OBS
Pacific chorus frog	<i>Pseudacris regilla</i>	5
American Bullfrog	<i>Lithobates catesbeianus</i>	3
Foothill Yellow-legged Frog	<i>Rana boylei</i>	3
Western Toad	<i>Anaxyrus boreas</i>	2
Rough-skinned Newt	<i>Taricha granulosa</i>	1
California Giant Salamander	<i>Dicamptodon ensatus</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment This site in Sonoma County, California exhibits a mixed oak woodland and grassland community with adjacent riparian or wetland features, as evidenced by the diverse herpetofauna and presence of both terrestrial and aquatic bird species. The wildlife community includes several species of regulatory concern: Foothill Yellow-legged Frog and California Giant Salamander indicate cool, clean flowing water habitats requiring protection; Northwestern Pond Turtle suggests pond or wetland resources; and the diversity of native reptiles and amphibians reflects relatively intact habitat corridors. The prevalence of ground-nesting birds (California Quail, Western Bluebird), grassland-associated mammals (Black-tailed Jackrabbit, Coyote), and the presence of sensitive amphibian species indicate moderate disturbance levels with functional open space, though the Eastern Fox Squirrel (non-native) suggests some prior habitat modification. Any site development should prioritize avoidance or mitigation of aquatic and riparian features, maintain wildlife movement corridors, and incorporate measures to protect breeding habitat for ground-nesting birds and reptile refugia. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳ PASSIVE + ACTIVE SOLAR PLANNING — Lat 38.35°

1 · AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 · IDEAL SOLAR ARRAY PITCH

Optimal tilt = 38.3° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97%	good
5/12	22.6°	97.6%	good
6/12	26.6°	98.2%	→ excellent
7/12	30.3°	98.8%	→ excellent
8/12	33.7°	99.3%	→ excellent
9/12	36.9°	99.8%	→ annual optimal
10/12	39.8°	99.8%	→ excellent
12/12	45°	99%	→ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	51	31	Minimize south glass
South glass (SHGC 0.6)	76	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	27	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	236	301	Highest load surface
Roof — light (absorb 0.35)	87	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	62	79	Best roof option for heat
South wall — dark	94	69	
South wall — light	31	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.1	1.9	2.4	OK
30ft	All year	3.1	2.9	3.7	Zone 2 check
40ft	All year	4	3.8	5.2	△ verify
50ft	All year	4.8	4.6	7.3	△ verify
60ft	All year	5.6	5.5	9.3	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 • ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🔍 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan Your location gets about 6.6 hours of useful sunlight per day on average, which is solid—think of it like having a reliable part-time job powering your home; in summer you'll get nearly 9 hours, but winter dips to under 4, so a solar system here needs to be sized with that winter reality in mind. Your roof's pitch is nearly perfect for solar panels, but here's what surprises most people: a dark roof absorbs three times more summer heat than a light-colored one, so if you're not going solar right away, a light roof color could cut your cooling costs dramatically—imagine the difference between wearing a black shirt versus a white one in the sun. The west side of your house is your biggest problem in summer, especially any windows facing that direction; afternoon sun pouring through west-facing glass is like opening your oven door in July, so shading those windows or using reflective film there pays back faster than almost any other improvement. Here's the good news: a large deciduous tree on the south side of your home is one of your best free tools—it'll block about 4 hours of intense summer sun while letting winter warmth through when the leaves drop, and it costs almost nothing compared to mechanical cooling. The difference between your current wall construction and something like insulated concrete forms (ICF) matters mainly because ICF holds onto heat or coolness longer—about 7 hours versus less than 1 hour—which means fewer temperature swings and less strain on your heating and cooling system. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.



Technical Brief — for architects, engineers, and surveyors

Site Intelligence Briefing ****38.34586°N, 122.30466°W | Napa County, California****

****Structural Design Drivers & Climate Envelope****

This site falls within IECC Zone 3C (Marine) with an Annual Freezing Index of 0°F-days, indicating negligible frost demand and mild winter temperatures year-round. Wind, snow, and seismic design parameters are not available in the current dataset, preventing determination of which load case governs structural design at this moment. Once wind speed (3-second gust) and seismic design category data are obtained from ASCE 7 maps and USGS tools respectively, the design team can rank load cases and select an appropriate lateral system. The marine climate classification suggests moderate thermal swings and potential salt-air exposure if proximity to the coast is relevant, though exact distance to tidal influence is not specified in the briefing data.

****Geotechnical Foundation Requirements****

The site soils are classified as Coombs gravelly loam with 2 to 5 percent slopes, rated Hydrologic Group C with well-drained characteristics. This drainage classification indicates moderate infiltration rates and low runoff potential—favorable for conventional shallow foundations and site stormwater management. Soil texture data is incomplete (marked unavailable), so bearing capacity and expansivity must be verified through Phase I subsurface exploration before footing design proceeds. The absence of frost depth data means local building code frost line requirements (typically 18–24 inches in Napa County) must be confirmed with the AHJ, as this will set minimum footing depth regardless of soil performance.

****Hazard Assessment & Subsidence****

No Quaternary faults appear within the search radius, and no subsidence basin data was detected, reducing two significant geotechnical risks for this Napa County location. Seismic hazard level cannot be assessed without design category assignment; however, Napa County's proximity to the San Andreas and associated fault networks warrants standard seismic evaluation during preliminary design. Wildfire hazard data are not present in the current briefing; defensible space and vegetation management requirements should be confirmed with county fire authority and CAL FIRE mapping tools if the site is in a Wildland-Urban Interface zone.

****Mining & Extraction Context****

Three active mineral producers operate within a 5-mile radius: Basalt Rock Co. (2 miles, clay operations), Mt George Pumice Deposit (3.1 miles), and Silva Pumice Quarry (5 miles). While none are in immediate adjacency, cumulative vibration from active blasting or heavy equipment traffic warrants confirmation of operational schedules and blast protocols with each operator, particularly if sensitive equipment, recording studios, or precision manufacturing will occupy the site. The broader mineral district (200 MRDS deposits within ~35 miles) reflects Napa's geology but poses no direct risk at this location. No oil or gas development is active nearby; the nearest plugged well is 7.2 miles away.

****Bottom Line:**** Obtain ASCE 7 wind speed and USGS seismic design category data immediately, confirm frost depth and full soil texture with the building department and Phase I exploration, and verify operational status and blast schedules of the three active quarries within 5 miles.



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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

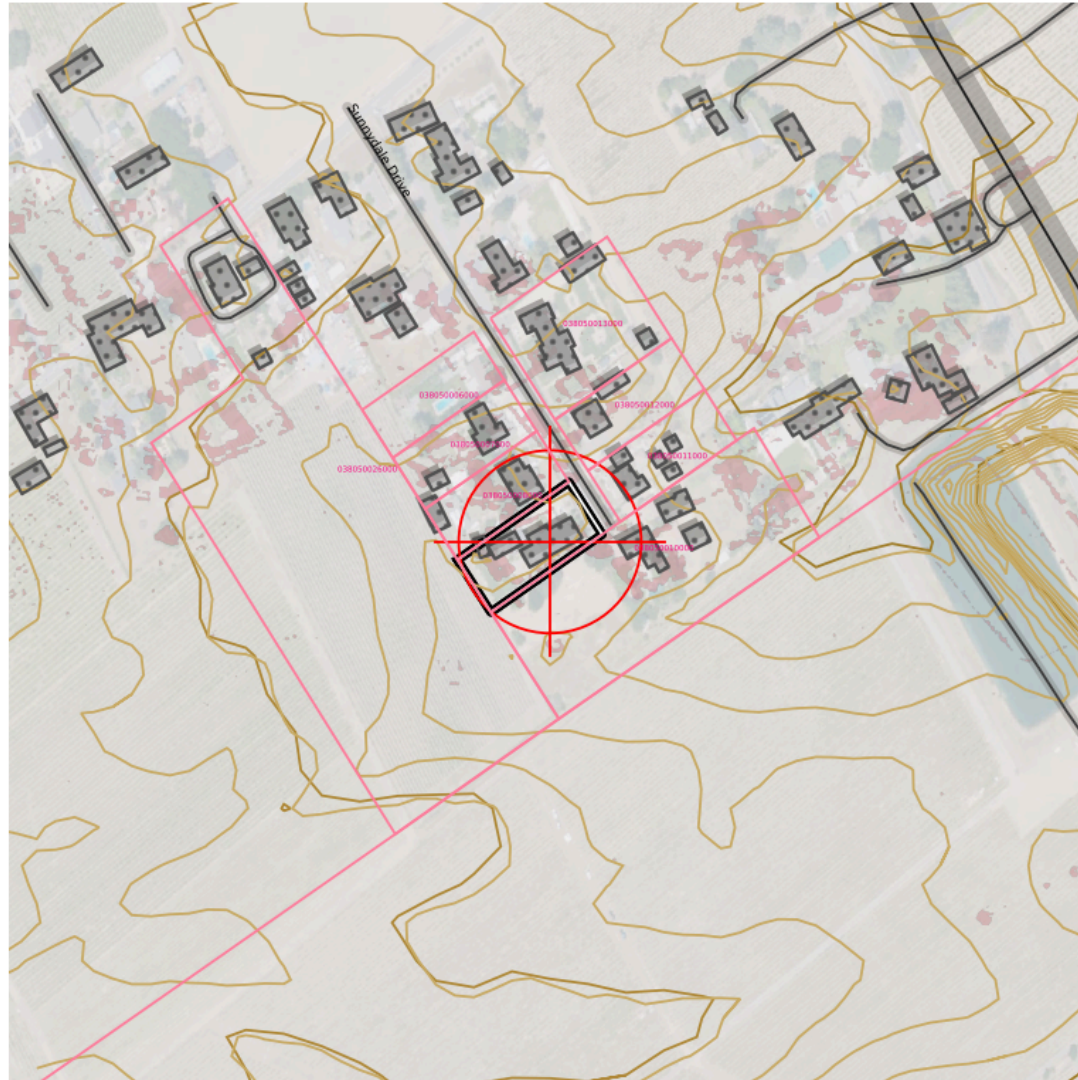


TOPOSCOUT
FIELD MAP

TopoScout Field Map
Lat: 38.34586 Lng: -122.30466
38.345860°N 122.304660°W | Radius: 30 mi

Generated: July 13, 2026
Base: USGS National Map
Generated by TopoScout - outkastdesigns.com

TOPOSCOUT SITE PLAN PREVIEW | \$19 UPGRADE FOR AUTOCAD DXF FILE



INCLUDED LAYERS



0 — 400 ft State Plane · NAD83 · US Feet

- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

